

Alexa Thinks Ahead

Built for the Next Billion.

AI that understands your home
before you ask.



THE NEXT TEN MINUTES

From a command-taker to a *household partner.*

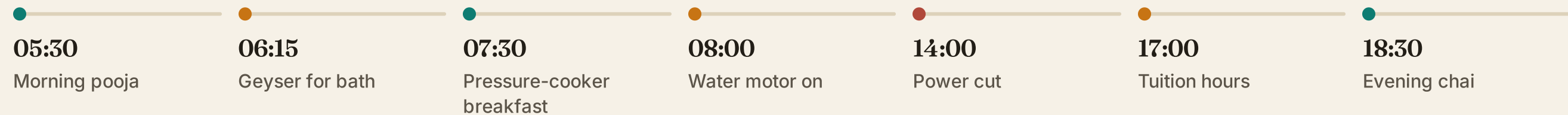
Alexa that learns the rhythm of an Indian home and acts **before** anyone asks — every action explained, every liberty earned.

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THE PROBLEM

Indian homes run on rhythm. Smart homes still **wait for orders.**

“Alexa, you should already know this.” — a multi-generational household issues dozens of commands a day to devices that never learn what happens next.



Reactive, not ready

"Alexa, turn on the geyser" — every single morning. The assistant obeys, but the hot water is still 20 minutes late.

Context-blind

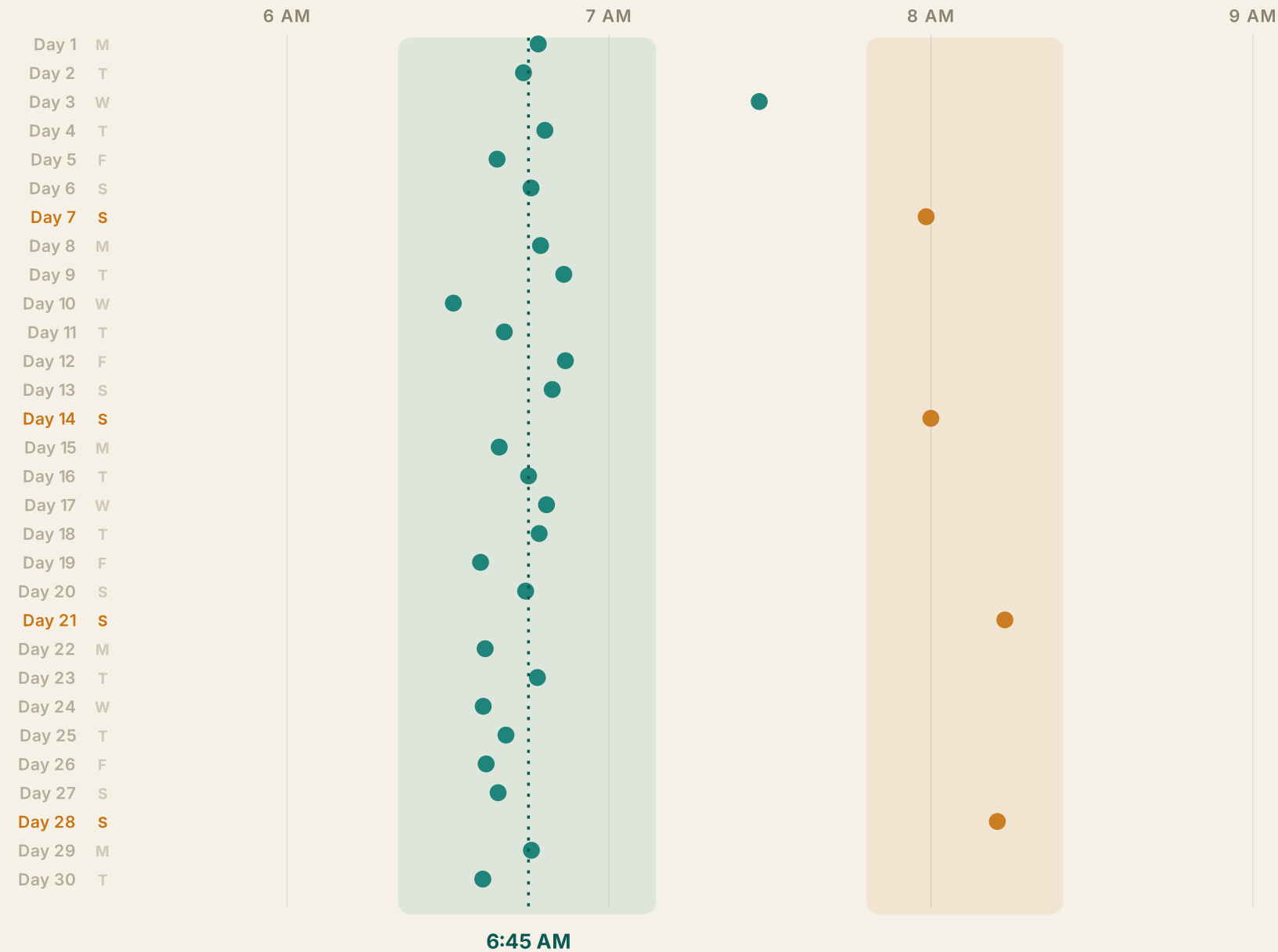
It doesn't know a power cut is different during a child's online class, or that grandparents bathe before everyone wakes.

Trust was never earned

No explanations, no way to correct it — so families keep automation switched off and do everything by hand.

THE SCENARIO — DADAJI'S BATH

30 mornings. *One pattern.*



Four things a family would *actually ask for*.

“It should already be ready.”

It acts ahead of the routine — the geyser is hot before the 6:45 bath, the living room is cool before Rajesh walks in. Zero commands for the daily rhythm.

“It should understand my home.”

India-first context, not hardcoded rules — pooja mornings, tariff windows, power cuts, tuition hours all ride into the reasoning as context it actually weighs.

“It should never overreach.”

Autonomy is graduated and earned — five trust tiers per device category; one override immediately lowers what it may do on its own.

“It should explain itself.”

Every action ships with a plain-language reason — "pre-heating the geyser at 05:15, the family wakes around 06:00" — and a one-tap Override.

And quietly, a fifth: **privacy** — the routines are learned from the home's own history, and they stay the home's.

LIVE DEMO

One day in an *Indian home*.

demo.mp4

drop the screen-recording into
deck/assets/

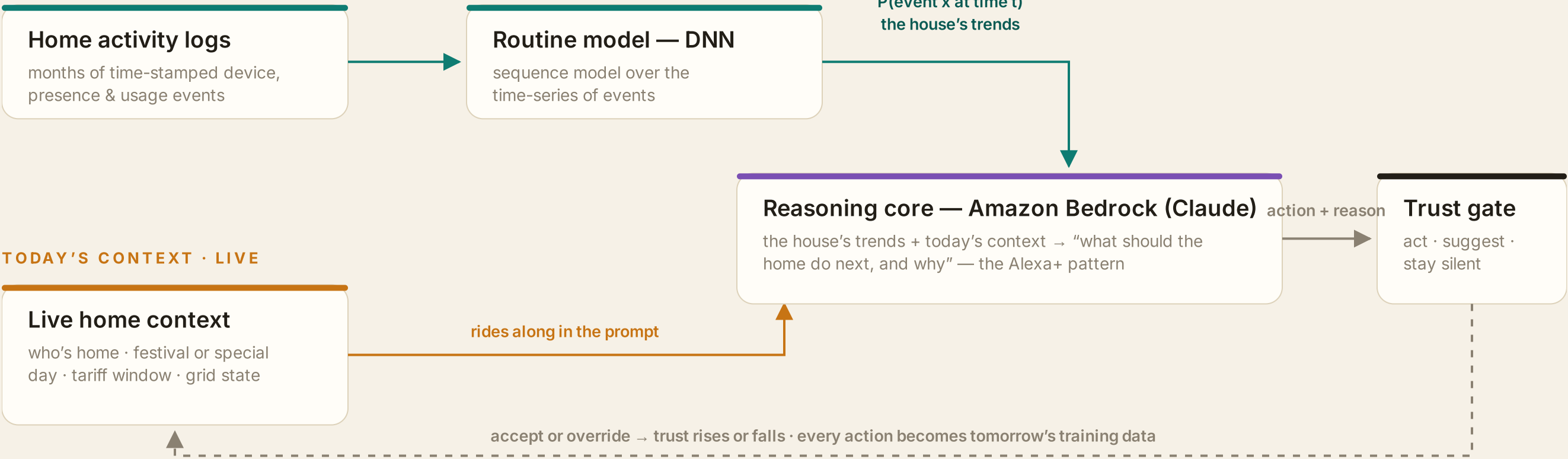
▶ 0:00



A routine model that knows the house, an LLM that knows *today*.

Two timescales, one decision: slow-learned trends meet live context inside the reasoning core.

LEARN THE RHYTHM · TRAINED ON THE HOME'S OWN HISTORY



Learning is separated from deciding

The sequence model distills months of time-series logs into event probabilities — cheap to run nightly, easy to retrain.

The LLM sees today, not just history

Occupancy, festivals, tariff windows and grid state ride along in the prompt — the same pattern Alexa+ uses for contextual reasoning.

Trust gates every action

Confidence and earned trust decide whether Alexa acts, suggests, or stays silent — and feedback flows straight back into the loop.

IMPACT & FUTURE VISION

Built for the *next billion* homes.

0

commands needed for the morning routine — hot water, lights and locks are simply ready

~30%

of flexible load shifted off peak tariff by anticipating usage, not reacting to it

100%

of proactive actions explained in plain language — with a one-tap Override

5

autonomy tiers per device category — families decide how much Alexa may do alone

WHERE THIS GOES NEXT

Sensor fusion

Hear the pressure-cooker whistle and the water-motor hum — context from sound and power signatures, no new hardware.

Hindi & Hinglish voice

Proactive suggestions in the language the household actually speaks, member by member.

Private by default

Routine learning moves on-device; only anonymised patterns ever reach the cloud.

Beyond one home

Matter ecosystem support and society-level insight: a colony that pre-arms for the 14:00 power cut together.

THANK YOU

Questions?

An Alexa that earns trust, one morning chai at a time.

Live demo — 54.252.207.144

github.com/piyushk-dev/hackon_project

Team **Bar Raisers** · HackOn with Amazon